

PROJECT AND TEAM INFORMATION

Project Title

CompressIt — File Compression, Backup and Search Utility Engine

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Managing many files can become difficult because they take up a lot of storage, duplicate files use unnecessary space, and making backups repeatedly can take time. searching for specific information in multiple text file can also be hard.

Our project, CompressIt, is designed to bring these tasks into one application. In it we'll provide file compression, duplicate-file backup, detection, and text searching in a single tool. we'll use Huffman Coding for compression, hashing for detecting duplicate/changed files, and an inverted index for faster text searching.

This project will also help us to understand how the Data Structures and OOP concepts that we are learning can be applied to a real application.

State of the Art / Current solution

There are quite a few tools available for file management. for example, you can use WinRAR to compress files . you can also have different backup apps that create copies of important files . some backup systems also support incremental backups, where only the changed files are saved.

There are also different search tools that can search through multiple files, or use indexes to make searching faster.

CompressIt is all about combining these features into one application. Our project will provide all of them together instead of different tools for compression, backup, duplicate detection and searching.

Also, we want to implement the main algorithms and data structures ourselves in C++, so that we can understand their practical application.

Project Goals and Milestones

Project Goals

Phase 1 - Planning and Designing

- *Complete the requirement and features of the project.*
- *Design the system architecture and plan the application of Data Structures and OOP*

Phase 2 – Developing

- *Develop the modules in C++ and file handling. the functionalities of Huffman compression, hashing, incremental backup, and inverted-index based searching.*

Phase 3 – Integrating and Testing

- *Integrate all the modules to form a utility.*
- *Test the whole system and assess the results like compression ratio, storage space saved, and search efficiency.*

Project Milestones

1. *Define the specifications, features, and the overall architecture of the system.*
2. *Implement the basics of file handling and management in C++.*
3. *Implement Huffman tree and compression/decompression operations.*
4. *Implement hashing operations for duplicate and modified file identification and incremental backup.*
5. *Build the inverted index and text search module.*
6. *Combine the modules and ensure that CompressIt is functioning properly with different types of files.*

Project Approach

We will design CompressIt incrementally in C++. at first, we will figure out the overall design and architecture of the program and split it into various modules like compression, backup, duplicate detection, and text search.

To compress the files, we will employ Huffman Coding algorithm with priority queue and binary tree. for comparison and detection of duplicates or modified files, we will use hashing. Backup feature of the system will make only changes to the files rather than backing up all files.

For searching, we will use hashing to build an inverted index for fast word search among files.

C++ will be the core language used in development of the program. STL, file handling and OOP concepts will be used while coding. in order to keep proper structure of the code, we will try to use classes such as Compressor, BackupManager, FileManager and Search Engine. at first, each module will be developed individually and then all modules will be brought together as one application to test and analyze. The compression ratio, storage saving from backup and search speed will be analyzed.

DATA STRUCTURE USED:

Binary Tree: In huffman compression, where the characters are kept in the huffman tree, paths from root to the characters give codes.

Priority Queue: In huffman compression, where two most frequent nodes are chosen continuously in order to create huffman tree.

Hash Table: In duplicate detection, where the hashes of the files are compared for detecting the duplicates files.

Hashing: In backup, where the hash of the present file is compared with the previous hash for detecting any changes.

Inverted Index: In text search, where the words are mapped with the files and their occurrences.

Vector: To store the information of the files, frequency of the characters and the search results.

Sorting: Used to arrange search results or file information on basis of frequencies, size or name.

Stack: This technique is used for history versioning where the backup version is the first one.

OOP CONCEPTS USED:

Classes and Objects: Used to break down the code into different modules like Compressor, BackupManager, FileManager and SearchEngine.

Encapsulation: It is used to hide the data and functions of a particular module in the class.

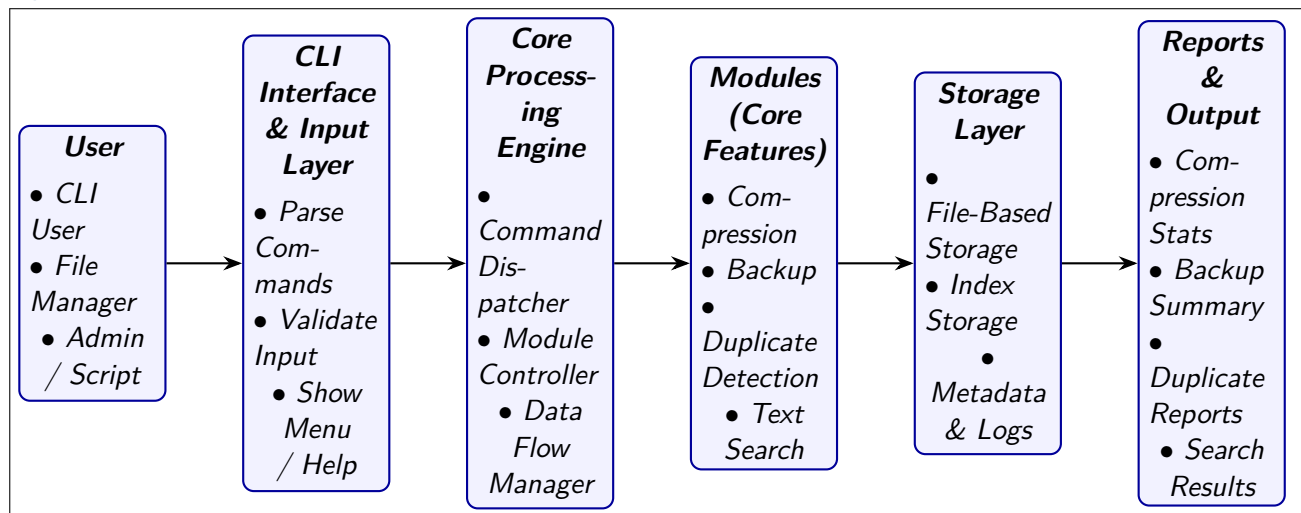
Abstraction: It is used to hide the compression process, backup process, and search process modules.

Constructors: Used to instantiate objects and their data.

STL: Used in C++ for data structures like priority queue and hash based containers.

File Handling: Used to read, write, compress, and restore files.

System Architecture



Project Outcome / Deliverables

The primary objective of the current project is the development of a C++ file management utility named as *CompressIt*. This utility will integrate various file management functions into a single utility application including compression and decompression of files using Huffman Coding, incremental backups, duplicate file detection through hashing, and searching for text across multiple files using inverted index. The application will contain a user-friendly command-line interface that allows users to take advantage of its various capabilities. In addition to this, after completing the development phase, testing of this application will be done based on some performance parameters including the compression ratio, savings due to backup, and search speed.

Assumptions

It is assumed that the user will deal mostly with standard file formats like text files, code files, log files, and documents.

The software will be running on the local machine and will need sufficient memory to make backups. It is assumed that the files are accessible, readable, and not password protected or corrupted.

For the first version, the software will concentrate on compression, backup, duplications check and text search.

References

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